

Team: Keyboard Gremlins

COS301 Capstone Project
University of Pretoria

Hackathon Platform

Admin Portal & Participant Portal User Guide



User Manual

Name	Student Number
Ayush Sanjith	23535424
Jasveer Ramdhaw	23537036
Rene Reddy	23558572
Heinrich Römer	23538181
Kwaku Asiedu (Sam)	18100156

Team Contact: keyboardgremlins@gmail.com

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1. Introduction

Welcome to the Hackathon Platform

The Hackathon Platform is a cloud-agnostic system purpose-built for running deterministic optimization hackathons. Participants generate an output from a set of provided inputs, rules, constraints, and levels, and an official solver grades that output to produce a score and the highest score wins. The platform gives organizers a highly configurable way to run events, and give participants a single place to register, form teams, submit solutions, and track their standing on a live leaderboard.

System Overview

Running an optimization hackathon manually is operationally heavy as someone has to distribute problem files, collect submissions, run the solver, and keep a leaderboard up to date, usually under time pressure as a deadline approaches. This platform standardizes that entire lifecycle:

Event setup → Registration → Team management → Submissions → Scoring → Leaderboard → Insights

The platform is split into two portals:

- Admin Portal: used by event organizers to configure events, manage problems and levels, run live operations during the event, and review insights.
- Participant Portal: used by hackathon participants to register for events, form or join teams, download problem resources, submit solutions, and track the leaderboard.

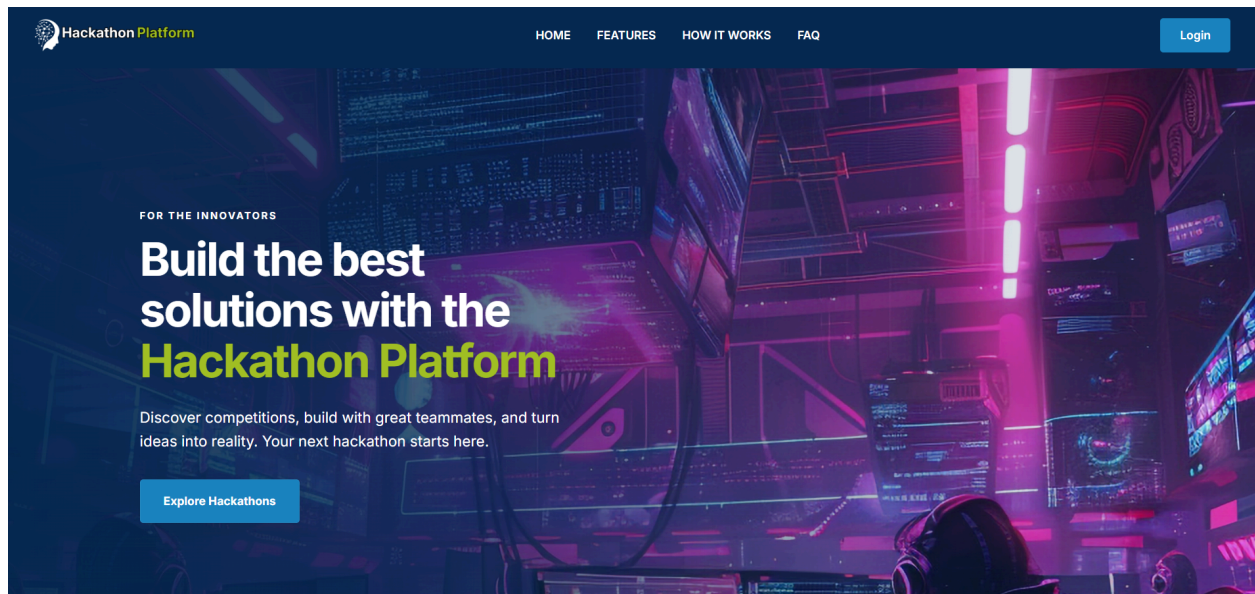
Document Purpose

This manual is your complete guide to using the Hackathon Platform, covering both portals. Whether you're the organizer setting up an event or a participant submitting a solution, this manual explains what each screen does and how to use it.

How to Use This Manual

- New Users: start with “Getting Started” and “Account Management”
- Admins: go to Part A - Admin Portal
- Participants: go to Part B - Participant Portal
- Part C : covers features shared by both portals, plus the Help and Support

2. Getting Started



What is the Hackathon Platform?

The hackathon Platform is a web application with two portals- an Admin Portal and a Participant Portal - built around a single core idea: organizers publish a problem, participants submit outputs against it, and an official solver scores every submission automatically and consistently.

Key Benefits

- Standardized event lifecycle: the same setup -> submissions -> scoring -> leaderboard flow works for any optimization challenge
- Consistent, automated scoring: every submission is graded by the same official solver, removing manual grading inconsistency
- Live visibility: participants see live leaderboard standing and a countdown timer; organizers see live event statistics
- Configurable events: public or private events, adjustable team sizes, dynamic levels, and controlled re-scoring after solver fixes

3. User Roles and Responsibilities

The hackathon Platform supports two primary user types, each with its own portal and permissions:

Admin(Event Organizer)

- Primary Role: creating, configuring, and operating hackathon events

Key Responsibilities:

- Create and configure events (name, schedule, team size, rules, visibility)
- Upload and maintain the problem statement, levels, resources, and solver
- Monitor registrations, team submissions, and platform health during the event
- Run live operations: extend time, freeze/unfreeze the leaderboard, push solver fixes
- Review statistics and insights to support decisions during and after the event

Participant

Key Responsibilities:

- Register for an event (and provide the event password/key for private events)
- Create a team or request to join an existing one
- Download and review the problem statement, rules, and level/resource files
- Submit solution outputs and source code archives for each level
- Review scoring feedback and logs, and track the live leaderboard

4. Account Management

4.1 Creating an Account



Create an account

Join the Hackathon Platform and start building great solutions.

First name
Enter your first name

Last name
Enter your last name

Email
Enter your email

Password
Enter your password

Confirm password
Re-enter your password

Password must contain:

- At least 8 characters
- One uppercase letter
- One lowercase letter
- One number
- One special character

Create account

or

 Continue with Google

Already have an account? [Log In](#)

Both Admins and Participants use the same account system, with access to each portal controlled by the account's assigned role.

1. Navigate to the platform's landing page and select "Login"
2. Complete the registration form:
 - Username: must be unique across the platform
 - Email Address: used as your login identifier
 - Password: choose a strong password
3. Submit the form. Your account is created and ready to use immediately

4.2 Logging In



Login

Enter your email and password to continue.

Email

Password

[Forgot password?](#)

Log in

or

 Continue with Google

Don't have an account yet? [Register](#)

4. Navigate to the login page.
5. Enter your registered email and password.
6. Select “Login.” You’ll land on the Admin Dashboard or the Participant Dashboard depending on your account’s role.

4.3 Profile Management

From the account menus you can view your account details, update notification preferences, and change your password.

4.4 Logging Out

Use the “Sign Out” option in the account menu to safely end your session, especially on a shared computer.

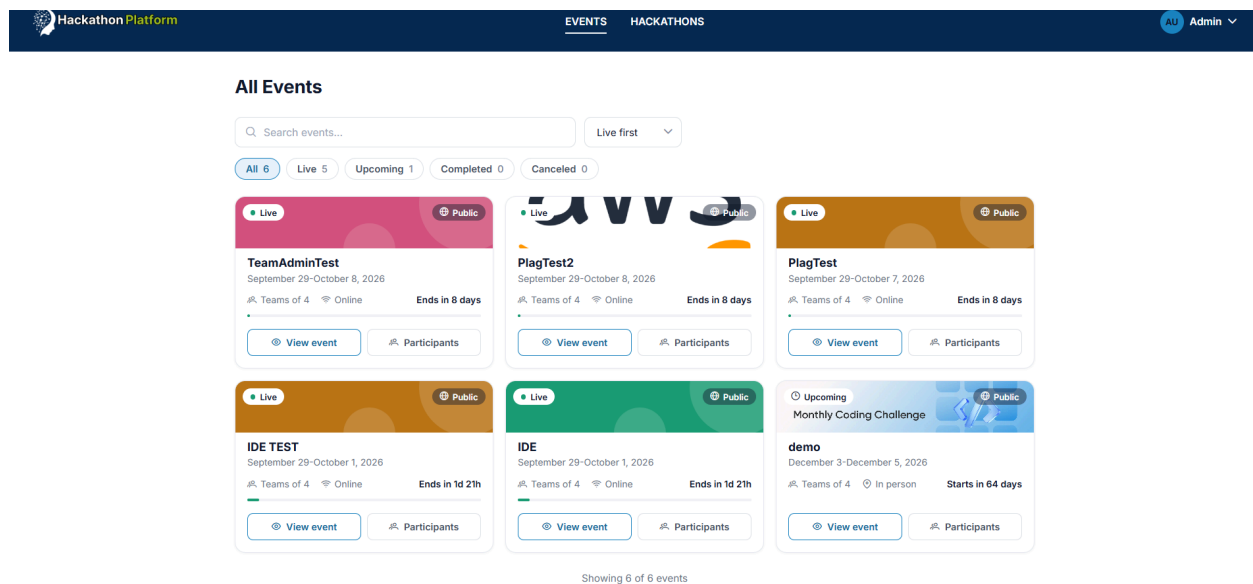
PART A

Admin Portal

5. Admin Portal Overview

The Admin Portal is organized into two layers. The Admin Event tab is the landing screen after logging in as an Admin and gives you a snapshot across every event you run. The Hackathons screen is the top-level list of the hackathons you organize (e.g “Entelect Challenge 2026”); opening a hackathon gives you its Levels, Solver, and Events tabs. Opening an individual event gives you that event’s own workspace, with tabs for Dashboard, Live Control, Announcements, Forum, Teams, Dietary Requirements, and Manage Event.

5.1 The Event Workspace

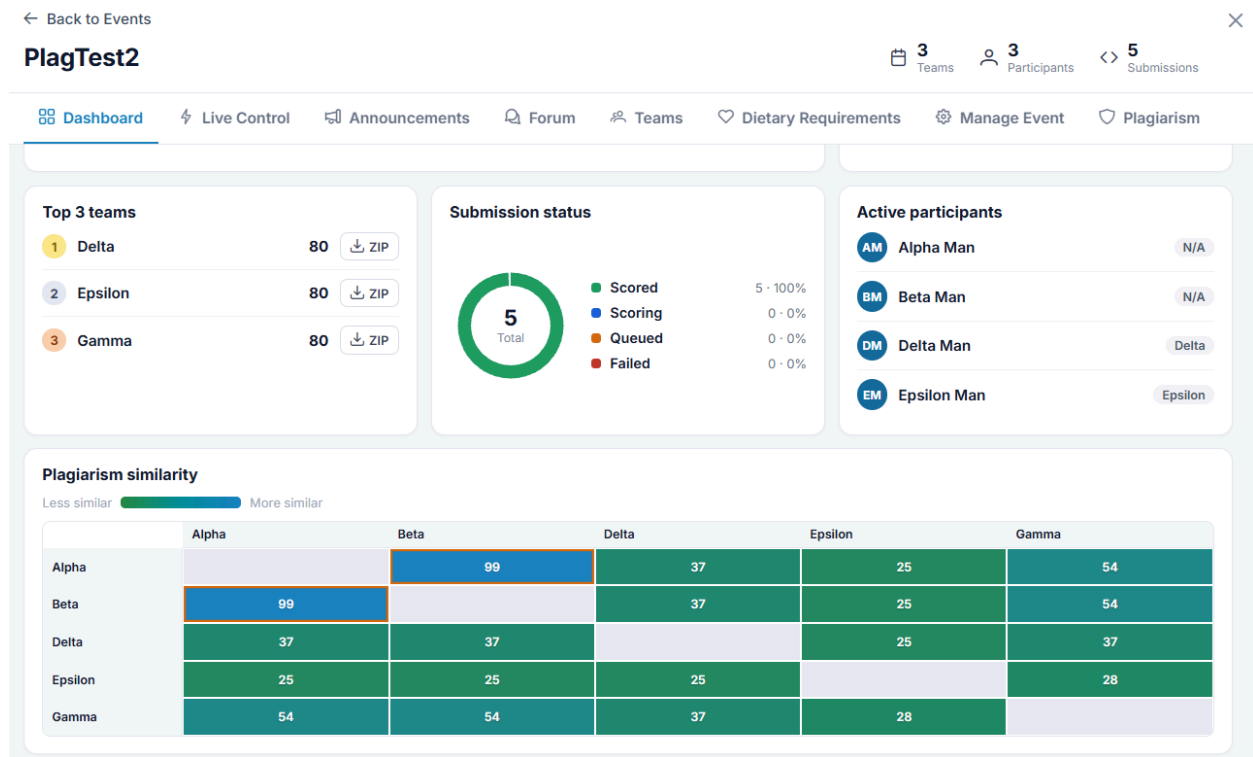


Opening an event - either from the Events tab, or the “View event” preview modal - brings up that event’s own workspace. A bar across the top shows a back button, the event’s name and status pill, and live counts of teams, participants, and submissions, followed by a row of tabs:

- Dashboard - live insights and charts for the event
- Live Control - extend the timer and pause/resume scoring
- Announcements - post updates to participants
- Forum - moderate discussion threads
- Teams - manage the event’s teams and members
- Dietary Requirements - review allergies and dietary needs
- Manage Event - edit the event’s core details

The same set of tabs is also available inside the “View event” preview modal opened from the Events grid, so you can check on an event without leaving that screen.

5.1.1 Event Dashboard



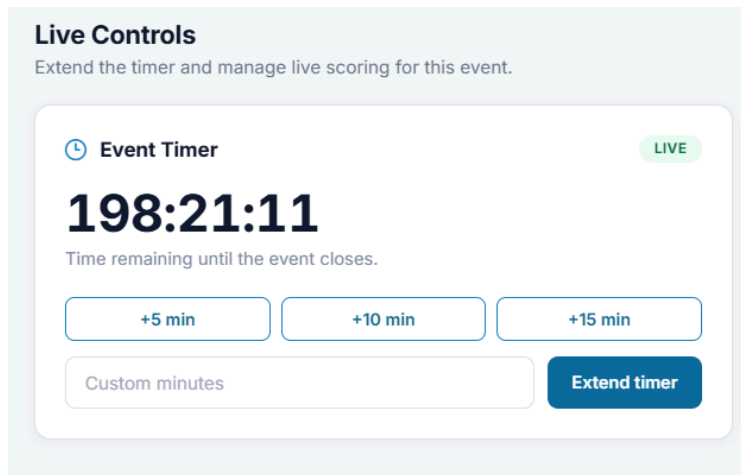
The Dashboard tab shows a customizable set of insight blocks. Select “Customize layout” to enter editing mode, where you can drag any block onto another to reorder them, or “Reset” to restore the default layout; select “Done” to exit editing mode.

Available blocks:

- Submission Trend - a line chart of submissions per minute over the last 60 minutes, with chips showing submissions in the last hour and the current error rate
- Top Teams - a ranked list of the leading teams by best score
- Submission Status - a donut chart breaking submissions down by status, with a legend showing counts and percentages
- Score by Level - a row per level showing the minimum, average, and maximum score, and how many submissions were scored
- Active Participants - a list of approved participants for the event and their teams

Each block shows its own loading or empty state (for example, “No submission activity in the last 60 minutes” or “No scored submissions yet”) until there is data to display.

5.1.2 Live Control

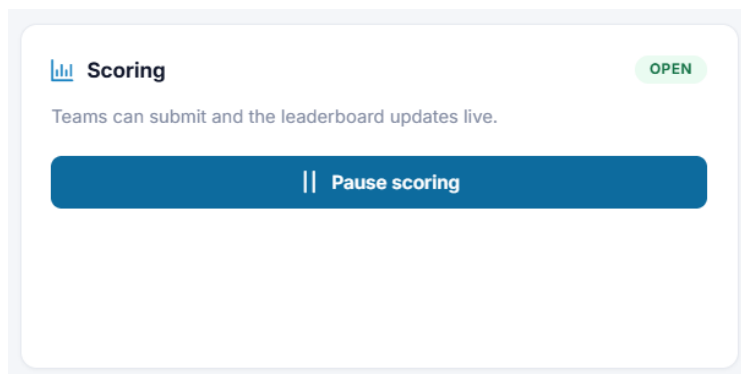


The Live Control tab lets you extend the event and manage live scoring.

Event Timer

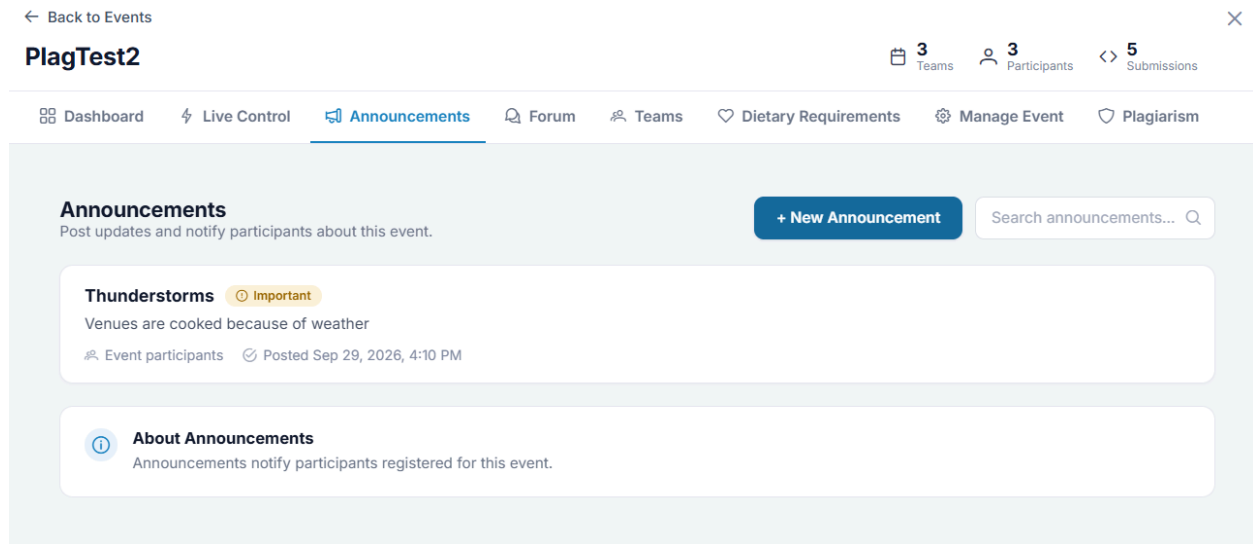
Shows whether the event is Live or Ended and the time remaining until it closes. Use the quick extend buttons(+5min, +10min, +15min) for common adjustments, or enter a custom number of minutes and select “Extend timer” for anything else. The buttons are disabled while an extension is in progress, and an error message appears if the extension fails.

5.1.3 Scoring



Shows whether scoring is Open or Paused. While scoring is open, teams can submit and the leaderboard updates live; while paused, submissions and leaderboard updates are put on hold for every team. Select “Pause scoring” or “Resume scoring” to toggle this.

5.1.4 Announcements



The Announcements tab is where you post updates and notify participants registered for the event. Use the search box to find past announcements,

Each announcement card shows its title, a severity badge (Info, Important, or Urgent, each with its own icon and color), the message body, and meta information showing who it was posted to and when. If none have been posted yet, the page shows “No announcements yet.”

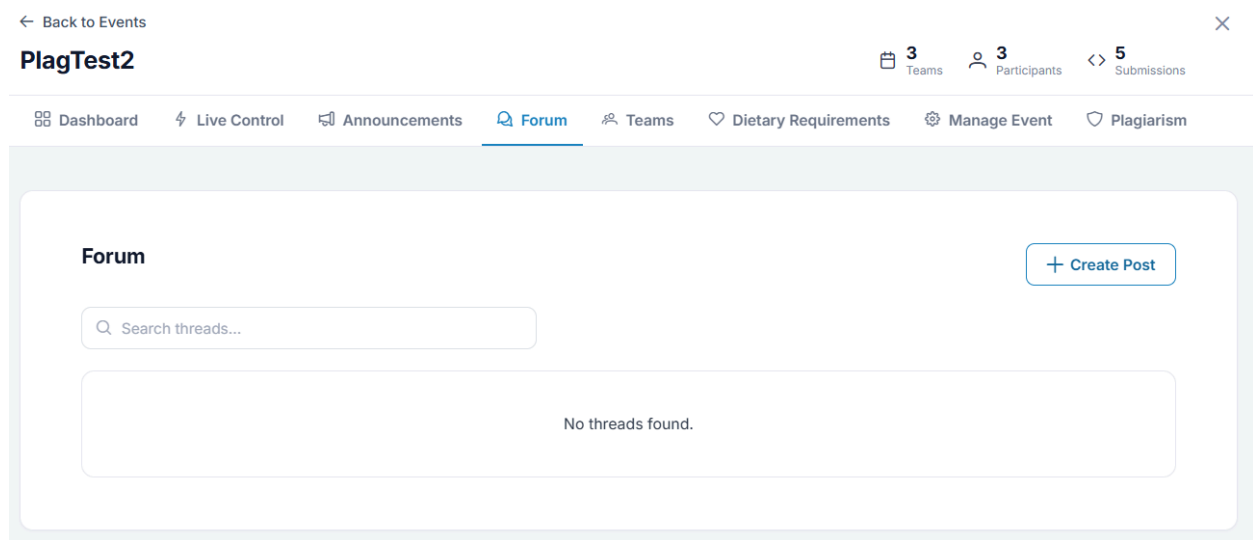
Posting an Announcement

Select “+ New Announcement” to open the announcement dialog.

1. Enter a Title.
2. Enter a Message describing what you want to tell the participants.
3. Choose a Severity: Info, Important, or Urgent.
4. Select “Post announcement” to publish it immediately to everyone registered for the event.

The button shows “Saving...” while the request is in progress. Select “Cancel” at any point to close the dialog without posting.

5.1.5 Forum



The Forum tab lets you moderate the discussion threads for an event. Use the search box to find threads by title.

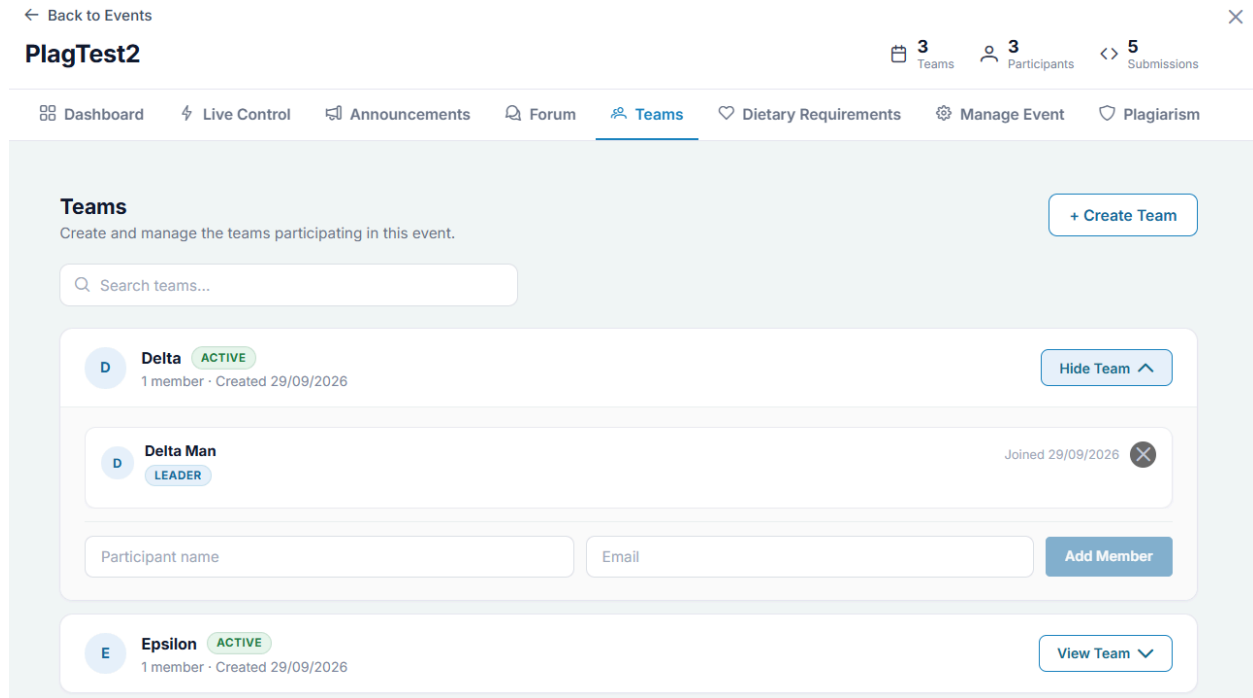
Each thread shows the author’s initial and role tag (Admin or Participant), its title, who started it, the number of replies, and when it was last active. Select “View Thread” to expand a thread and read its root message and replies, or “Hide Thread” to collapse it again.

Moderation and replying

- A delete icon appears next to any message you are permitted to remove (your own messages, or any message if you have moderator permissions) - use it to delete a message from a thread
- If you have comment permissions a reply box appears at the bottom of an expanded thread; enter your reply and select “Post Reply” (disabled until you type something)
- If you have permission to create posts, a “+Create Post” button appears in the page header, opening a dialog to enter a Title and Body and select “Create Post”

If no threads exist yet, or none match your search, the page shows “No threads found.”

5.1.6 Teams



The Teams tab lets you create and manage the teams participating in an event. Use the search box to find a team by name.

Each team card shows its name, an Active or Banned status tag, its member count, and when it was created, along with a delete (trash) icon and a “View Team”/“Hide Team” toggle to expand its member list.

Managing members

Inside an expanded team, each member is listed with a role tag (Leader or Member) and, if applicable, a “Banned” tag. For each member you can:

- Ban or Unban the member
- Remove the member from the team entirely

At the bottom of the member list, enter a participant name and email and select “Add Member” to add someone directly to the team.

5.1.7 Dietary Requirements

← Back to Events

PlagTest2

3 Teams 3 Participants 5 Submissions

Dashboard Live Control Announcements Forum Teams **Dietary Requirements** Manage Event Plagiarism

Dietary Requirements

Review allergies and dietary needs across every team in this event.

PARTICIPANTS **5** WITH REQUIREMENTS **0** TEAMS **4**

Search participants or teams... Only show requirements ☐

N No Team 2 members

A Alpha Man None specified

B Beta Man None specified

D Delta 1 member

D Delta Man None specified

The Dietary Requirements tab lets you review allergies and dietary needs across every team registered for an event.

Stats at a glance

- Participants - total participants in the event
- With requirements - how many participants have specified a dietary requirement or allergy
- Teams - total number of teams

Use the search box to find participants or teams by name, and the “Only show requirements” toggle to hide participants with no requirements specified.

Participants are grouped by team, each shown with the team's avatar, name, and member count. Within a team, every member is listed with their name and their dietary tags (for example, allergies or dietary preferences); a member with nothing specified is shown as "None specified." If no participants match your search or filter, the page shows "No participants match your search."

5.1.8 Manage Events

[← Back to Events](#)

PlagTest2

3 Teams

3 Participants

5 Submissions

[Dashboard](#)
[Live Control](#)
[Announcements](#)
[Forum](#)
[Teams](#)
[Dietary Requirements](#)
[Manage Event](#)
[Plagiarism](#)

Manage Event

Edit event details and manage resources

Basic Information

Event name *

Event Branding

Event banner

Click to upload or drag and drop
PNG, JPG, JPEG or Webp
5:1 aspect ratio - 2000 x 400 px recommended
Current banner available

Event logo

Click to upload or drag and drop
PNG, JPG, JPEG or Webp
2:1 aspect ratio - 1600 x 800 px recommended
Current logo available

Start date & time *

Duration (hours)*

Description

Tagline

Rules

sdc v

Visibility & Access

Public
Anyone can register

Private
Registration key required

Team size limit *

Allowed Technology

Leave empty to allow any technology, or add specific stacks to restrict submissions.

Prizes
Add the prizes participants can win.

1	1st Place	R0	
2	2nd Place	R0	
3	3rd Place	R0	

+ Add prize

Event Configuration

Event format

Online
Participants can join remotely

In Person
Participants attend at a physical location

IDE

IDE Required
Participants use the platform IDE

External IDE
Participants can use their own tools

Cancel

Save Changes

The Manage Event page allows event organizers to edit and update the details of an existing event. From here, you can change the event's name, start date, and time duration, and description. You can also control who can access the event by setting its visibility to Public (anyone can register) or Private (a registration key is required), and set a team size limit for participants. Additionally, you can manage the leaderboard by keeping it Live so rankings update in real time, or Frozen so rankings are locked and hidden from updates. Once you have made your changes, click “Save Changes” to apply them, or “Cancel” to discard them. If you need to remove the event entirely, click “Delete Event”.

5.1.9 Participants

TeamAdminTest - Participants ×
Everyone registered for this event.

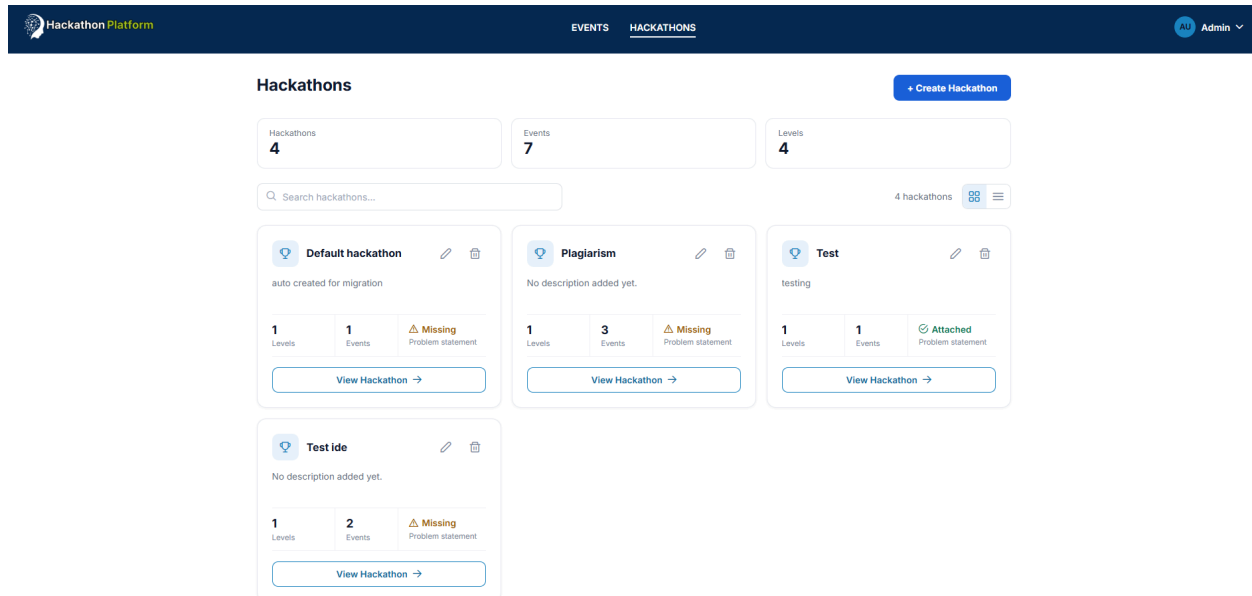
Search by name, email, or team... 2 of 2

PARTICIPANT	EMAIL	TEAM	ROLE	JOINED	ACTIONS
AM Alpha Man	a@hackathon.com	ALPHA	LEADER	Sep 29, 2026	Report Ban
BM Beta Man	b@hackathon.com	ALPHA	MEMBER	Sep 29, 2026	Report Ban

Select “Participants” on an event card in the Events grid to open the Participants modal for that event - a list of everyone approved into a team for it. Use the search box to filter by name, email, or team; a counter shows how many participants match out of the total.

The participants table lists each person’s name, email, team , role (Leader or Member), and the date they joined.

5.2 The Hackathon Workspace

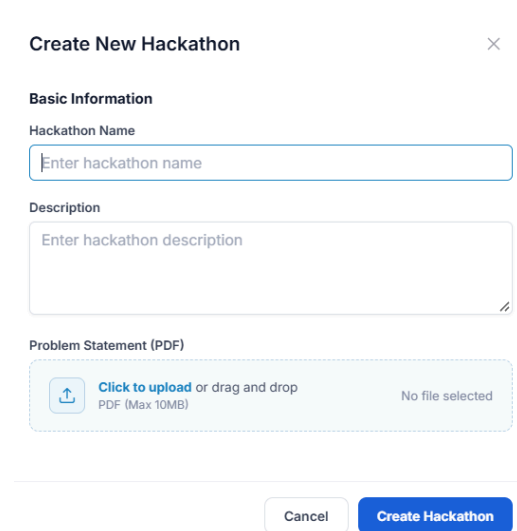


The Hackathons screen lists every hackathon you organize, and is the starting point for creating events and drilling into levels or the solver. A summary bar at the top shows the total number of hackathons, events, and levels across the platform.

Grid and list views

Use the search box to filter hackathons by name, and the view toggle in the toolbar to switch between a card Grid view and a compact List view. Both views show the same information.

1. Creating a Hackathon



The dialog box is titled "Create New Hackathon" with a close button (X) in the top right corner. It contains three sections: "Basic Information" with a "Hackathon Name" text input field and a "Description" text area; "Problem Statement (PDF)" with a file upload area showing "Click to upload or drag and drop", "PDF (Max 10MB)", and "No file selected"; and a bottom section with "Cancel" and "Create Hackathon" buttons.

Create New Hackathon

Basic Information

Hackathon Name

Enter hackathon name

Description

Enter hackathon description

Problem Statement (PDF)

Click to upload or drag and drop
PDF (Max 10MB) No file selected

Cancel Create Hackathon

1. Select "+ Create Hackathon". This opens the Create/Edit Hackathon dialog.
 2. Enter the Hackathon Name (required).
 3. Add a Description.
 4. Optionally upload a Problem Statement as a PDF (max 10MB) by clicking the upload area or dragging a file onto it. A success or error message appears once the upload completes.
 5. Select "Create Hackathon" to save. A confirmation toast appears in the top-right corner, and the new hackathon appears as a card.
- Select "Cancel" at any point to close the dialog without saving.

2. Editing a Hackathon

Edit Hackathon

×

Basic Information

Hackathon Name

Default hackathon

Description

auto created for migration

Problem Statement (PDF)

 Click to upload or drag and drop
PDF (Max 10MB)

No file selected

Cancel

Save Changes

Select the pencil (edit) icon on a hackathon's card or row to reopen the same dialog, prefilled with its current name, description, and problem statement, so you can update them. If a problem statement is already uploaded, a hint tells you a new file will replace it.

3. Hackathon Card and Row Actions

Each hackathon card (or row, in List view) shows its name, description , and three stats: Number of Levels, Number of Events, and whether a Problem Statement is Attached or Missing, along with these actions:

- View Hackathon: opens the hackathon , landing on its Levels tab
- Edit (pencil icon): opens the Create/Edit dialog to rename, update the description, or replace the problem statement
- Delete (trash icon): permanently removes the hackathon

If you haven't created any hackathons yet, the page shows "No hackathons created yet" with a shortcut to create your first one.

5.2.1 Navigating a Hackathon

Once you open a hackathon card shows its name, description, and level/event counts, followed by a row of tabs that stay in places as you move between areas of that hackathon:

- Levels - manage the hackathon's levels and their files
- Solver - upload and manage the official solver
- Events - manage the events that run under this hackathon

Select “Back to Hackathons” at any time to return to the Hackathons list.

5.2.2 Levels & Files

← Back to Hackathons

Default hackathon
auto created for migration

1 Levels 1 Events

Levels Solver Events

Levels
Drag and drop levels to reorder them.

+ Create Level Search levels... 🔍

ORDER	LEVEL	DESCRIPTION	RESOURCES	ACTIONS
1	Level 1: Phase One	No description	Starter ZIP	View Level Edit Delete

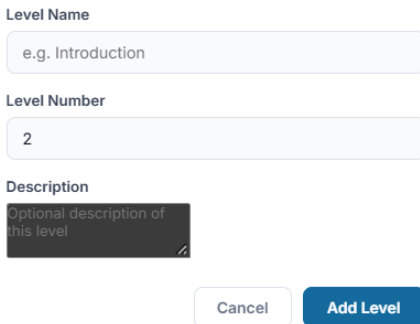
The Levels screen lists every level for the hackathon in a table, in order, and lets you drag and drop rows to reorder them - a banner reading “Saving new order...” appears while the new order is saved. Select “+ Create Level” to add another. If no levels exist yet, the table shows “No levels created yet”.

Each row shows a drag handle, the level's order number, its name, description, and how many resource files are attached, plus three actions:

- View Level: opens the Manage Files dialog for that level
- Edit (pencil icon): opens the level in the Add/Edit Level dialog
- Delete (trash icon): removes the level

1. Adding and Editing Levels

Add New Level



The 'Add New Level' dialog form consists of three input fields and two buttons. The first field is 'Level Name' with a placeholder 'e.g. Introduction'. The second field is 'Level Number' with the value '2'. The third field is 'Description' with a placeholder 'Optional description of this level' and a small icon in the bottom right corner. Below the fields are two buttons: 'Cancel' and 'Add Level'.

Level Name
e.g. Introduction

Level Number
2

Description
Optional description of this level

Cancel Add Level

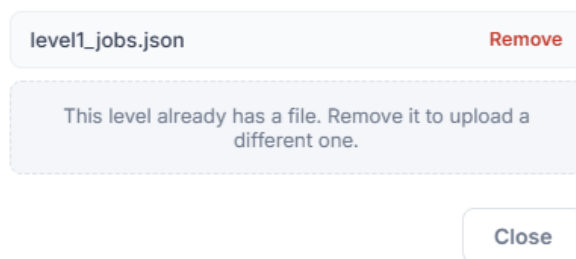
The Add/Edit Level dialog captures:

- Level Name - for example, “Introduction”
- Level Number - the level’s position/order
- Description - an optional description of the level, shown to participants

When editing an existing level, a “Delete Level” button is also available inside the dialog. Select “Cancel” to close without saving, or “Add Level”/“Save changes” to confirm.

2. Managing Level Files

Manage Files – Phase One



The 'Manage Files – Phase One' dialog shows a file 'level1_jobs.json' with a 'Remove' button next to it. Below this is a message box that says 'This level already has a file. Remove it to upload a different one.' At the bottom is a 'Close' button.

level1_jobs.json Remove

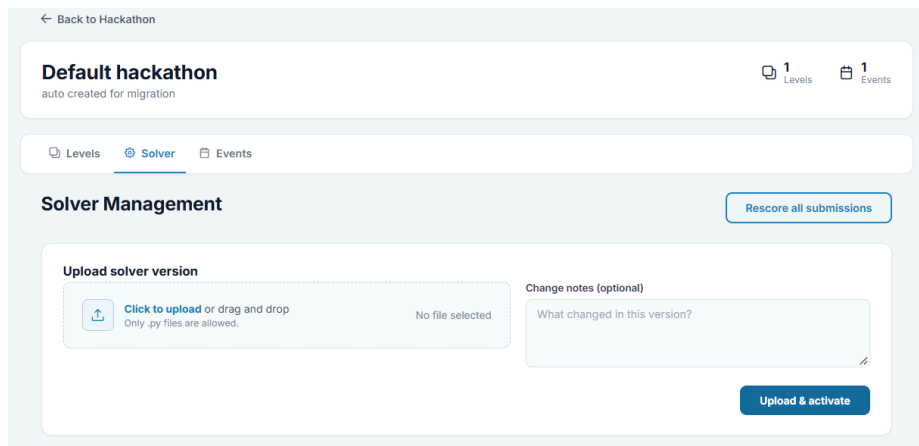
This level already has a file. Remove it to upload a different one.

Close

The Manage Files dialog lists the file currently uploaded for a level with a “Remove” button, or “No files uploaded for this level” if none exists yet. Each level currently supports a single file: once a file is uploaded, a note explains that the existing file must be removed before a new one can be uploaded, a note explains that the existing file must be removed before a new one can

be uploaded. Click the drop zone or drag a file onto it to upload. Select “Close” when you’re done.

5.2.3 Solver Management



The Solver page is where you upload and manage the official solver that grades all participant submissions for the hackathon. The solver is a Python script that takes a participant’s output file, the level input files, and the level number, and returns a JSON result with the score and validation messages.

Rescore submissions

Select “Rescore all submissions” in the page header to re-run existing submission through the currently active solver - useful after fixing a bug in the solver. The button shows “ Queuing rescore...” while the request is in progress, and a success or error message appears below the header once it completes.

1.Uploading a New Solver Version

The upload card accepts a solver file via drag-and-drop or by browsing. Only .py files are allowed.

1. Click the drop zone or drag a .py file onto it. The selected file name appears once chosen.
2. Optionally add Change Notes describing what changed in this version.

3. Select “Upload & activate” to upload the file and make it the active solver used for scoring.

The button shows “Uploading...” while the request is in progress, and a success or error message appears below the form once it completes. Only one solver version can be active per hackathon at a time; uploading a new version deactivates the previous one, but submissions already scored keep their original results until you choose to rescore them.

Solver Requirements:

- The solver must be a Python script (.py file)
- The script must accept exactly 3 command-line arguments in the following order:
 1. `output_file` - Path to the participant's submission output file
 2. `level_inputs_dir` - Path to the directory containing the level's input files
 3. `level_number` - The numeric identifier of the level being scored

Solver Output Format:

The solver must output a single line of valid JSON as its final line of output. The JSON must follow this schema:

```
{  
  
  "score": 85.5,  
  
  "status": "SCORED",  
  
  "errorType": null,  
  
  "messages": [  
  
    "Validation passed for all constraints",  
  
    "Output format is valid"  
  
  ]  
}
```

```
}
```

JSON Fields:

- **score** (number, required) - The numeric score for this submission. Higher scores indicate better performance. Must be a decimal number (e.g., 85.5).
- **status** (string, required) - Must be either **"SCORED"** or **"FAILED"**. If the solver encounters any validation errors or issues, return **"FAILED"**.
- **errorType** (string, optional) - If status is **"FAILED"**, this should indicate the type of error (e.g., **"MALFORMED_OUTPUT"**, **"RUNTIME_ERROR"**, **"INVALID_FORMAT"**). Can be **null** for successful runs.
- **messages** (array of strings, optional) - A list of human-readable messages providing feedback to the participant. Can be empty for successful runs.

Solver Environment:

- The solver runs in a temporary isolated directory
- The script is executed with the Python interpreter configured by the platform

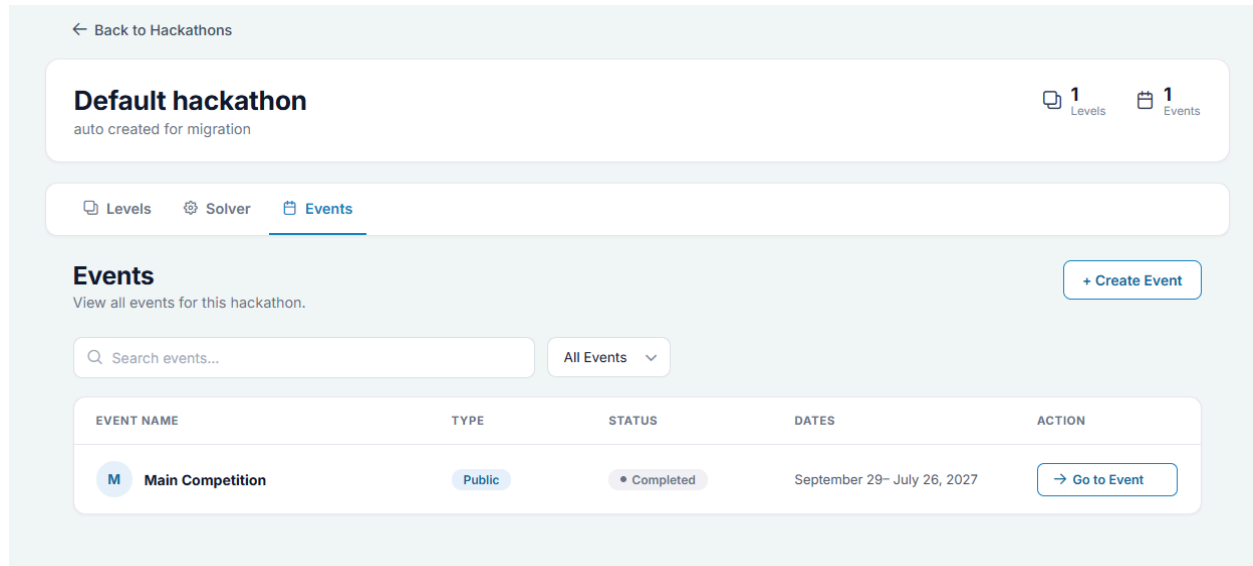
Upload Process:

1. Alongside the file, provide:
 - **Change Notes** - A short description of what changed in this version (optional)
2. Select "Upload & Activate" to upload the file and make it the active solver used for scoring

Important Notes:

- Only one solver version can be active per hackathon at a time
- Uploading a new solver version automatically deactivates all previous versions
- All submissions submitted before a solver update will retain their original scores
- To re-score submissions with the new solver, use the admin rescore functionality in the scoring operations

5.3 Managing Events



1. Viewing & Filtering Events

The Events tab, within a hackathon, lists every event created under it. Search events by name using the search box, sort them using the sort dropdown, and filter by status using the row of status chips (each showing a live count) above the list.

Each event is shown as a card with its banner and logo (or a coloured placeholder if none is set), a status badge (Live, Upcoming, Completed, or Canceled), and a visibility badge (Public or Private). Below that, the card shows the event's name, dates, configured team size, whether it is in-person or online, and - for live events - a progress bar showing how far through the event window you are. Two actions are available on every card:

- View event: opens a quick-preview modal of the event's workspace without leaving the Events screen
- Participants: opens the Participants modal for the event

While events are loading the page shows skeleton placeholder cards; if none match your search or filters it shows "No events match your search" (or an equivalent empty-state message). A "Showing X or Y events" counter and a "Load more" button sit below the grid when there are more events than currently displayed.

2. Creating a New Event

Create Event

Basic Information

Event Name

Enter event name

Event banner

Click to upload or drag and drop
PNG, JPG, JPEG or Webp
5:1 aspect ratio - 2000 x 400 px recommended

Event logo

Click to upload or drag and drop
PNG, JPG, JPEG or Webp
2:1 aspect ratio - 1600 x 800 px recommended

Description 0/1000

Describe your event...

Rules 0/2000

Enter the rules and requirements for participants...

Tags & Technology

Help participants find and prepare for this event.

Tagline

e.g. Build, compete, and innovate

A short one-line description shown alongside the event.

Allowed Technology

e.g. React, Python, Java

Leave empty to allow any technology, or add specific stacks to restrict submissions.

Schedule

Start Date

yyyy/mm/dd

Start Time

--:--

Duration (hours)

48

Team Member Limit

4

☒ The event will end automatically based on the start date, time and duration.

☐ This is an in-person event

☐ Use IDE for submissions

Leaderboard Freeze

yyyy/mm/dd --:--

Leaderboard rankings will stop updating publicly after this time.

Prizes

Add the prizes participants can win.

Access

Choose who can access this event.

Public

Anyone can discover and participate in this event.

Private

Only invited users or authorized participants can access this event.

Select “+ Create Event” from the Events tab to open the Create Event form. Select the close icon or “Cancel” at any time to return without finishing.

Basic Information

- Event Name - required
- Event banner - an image upload with a recommended 5:1 aspect ratio (2000 x 400px), PNG/JPG/JPEG/WebP
- Event logo - an image upload with a recommended 2:1 aspect ratio (1600 x 800px), PNG/JPG/JPEG/WebP
- Description - a free-text field with a live character counter
- Rules - a free-text field, also with a live character counter

Tags & Technology

- Tagline - a short one-line description shown alongside the event
- Allowed Technology - add technology chips by typing a name and pressing Enter or comma; leave empty to allow any technology, or add specific stacks to restrict submissions

Schedule

- Start Date and Start Time - required
- Duration (hours) - required; the event ends automatically based on the start date/time and duration
- Team Member Limit - required, minimum 1
- This is an in-person event - checkbox to mark the event as in-person rather than online
- Leaderboard Freeze - an optional date/time after which public leaderboard rankings stop updating

Prizes

Add one or more prizes, each with a Title (for example “1st Place”) and a Description (for example, “R10,000 cash + mentorship”). Select “+ Add prize” to add a row, or trash icon to remove one.

Access

Choose one of two access options:

- Public - anyone can discover and participate in this event

- Private - only invited users or authorized participants can access this event; selecting this reveals a required Registration Key field that participants must event to register

Submitting

Select “Cancel” to discard the form, or “Create Event” to finish. The button shows “Creating...” and is disabled while the request is in progress. If something goes wrong, an error message appears at the top of the form

3. Managing an Event

The Manage Event tab is where you edit an event’s core details after it has been created. While details are loading, the page shows “Loading event details...”; a success or error message appears above the form as you make changes.

Basic Information

- Event name - required
- Start date & time - required
- Duration (hours) - required
- Description - the event’s rules and description

Visibility & Access

The same Public/Private toggle as event creation - Public (“Anyone can register”) or Private (“Registration key required”), with a Registration Key field shown when Private is selected. A Team size limit field is also set here.

Leaderboard

A live/Frozen toggle: Live keeps rankings updating in real time, while Frozen locks the leaderboard and hides it from further updates.

Deleting an Event

Select “Delete Event” in the page header to open a confirmation dialog. Deleting an event permanently removes its registrations, teams, submissions, and uploaded resources, and cannot be undone; confirm by selecting “Yes, Delete Event”, or “Cancel” to back out.

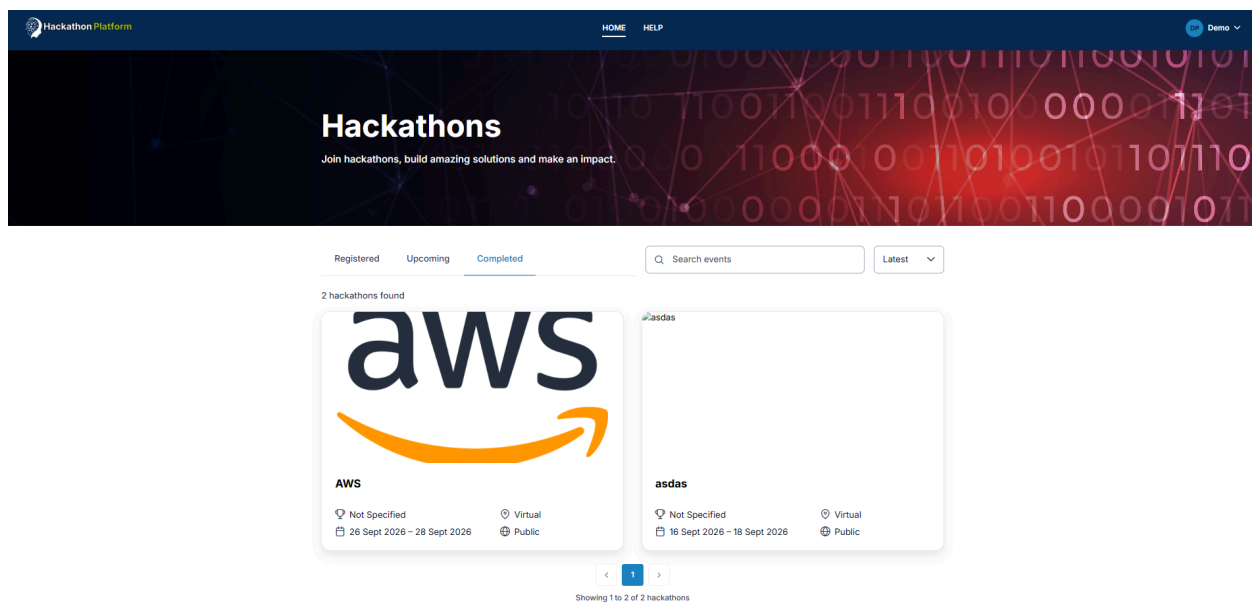
Saving

Select “Cancel” to discard changes and go back, or “Save Changes” to save. The button shows “Saving...” while the request is in progress.

PART B

Participant Portal

6. Participant Portal Overview



Before logging in, participants land on the public homepage, with a top navigation bar linking to Home, Features, How it Works, and FAQ, plus a Login button. Once logged in, participants browse events from the home page, register for the ones they want to enter, and then work inside the event's own workspace - a tabbed area with Overview, Rules, Announcements, Forum, Submission, Submission History, My Team, and Leaderboard.

6.1 Browsing Events



Events you can see - whether you're already registered or the event is still upcoming - are shown as cards. Each card shows:

- The event's logo image
- Its name and tagline
- Its total prize pool
- Whether its In-Person or Virtual
- Its dates
- Its visibility - Public or Private

An upcoming event also shows a countdown tag with the number of days until it starts. An event you're already registered for that is currently running shows a "Live" tag. Selecting a card opens the event.

6.2 Event Details & Registration

[← Go Home](#)

AWS

Download Certificate

Download Specifications

Opening an event brings up its banner image, name, and tagline, along with:

- A countdown timer (unless the event has already completed), broken down into days, hours, minutes, and seconds
- A primary action button - “Register” if you haven’t joined yet (becomes “Registered” once you have), or “Download Certificate” once the event has completed
- “Download Specifications” - downloads the event’s problem statement

Below that, a details strip shows the event’s Status, Prize Pool, Visibility, and Format (In-Person or virtual), plus while the event hasn’t finished - either “Submissions Open” (for an upcoming event) or “Submissions Close” (for an active event), with the exact date, time, and time zone.

Registering for an Event

Selecting “Register” opens a Register for Event dialog showing the event’s name, plus:

- For a Private event, a note that registration key is required, and a field to enter it
- For a Public event, a simple “Ready to Register?” confirmation
- For an in-person event, an additional note asking for any Dietary Requirements and Allergies, with a field for each

Select “Cancel” to back out, or “Register” to confirm - the button shows “Registering...” while the request is in progress.

6.3 The Event Workspace

Once you're inside an event, a row of tabs lets you move between its different areas without losing your place:

- **Overview** - the event's summary, key details, and prizes
- **Rules** - the event's rules and requirements the organizer has set
- **Forum** - discussion threads for the event
- **Submissions** - upload your solution for each level
- **Submission History** - your past submissions, scores, and logs
- **My Team** - create, join, or manage your team for the event
- **Leaderboard** - live standings for the event

Each tab loads its own content within the event workspace, so you can move between them without losing your place.

6.3.1 Overview Tab

← Go Home

AWS

Download Certificate

Download Specifications

Status	Prize Pool	Visibility	Format
Completed	R 0	Public	Virtual

Overview Rules Announcements History Rankings

Event Overview
Everything you need to know about this event.

About the Event

Event Details
Key information about participation.

Team Size Up to 4 members	Format Virtual
Visibility PUBLIC	Allowed Technologies java python

Prizes
Rewards for the top performing teams.

1st Place R0	2nd Place R0	3rd Place R0
-----------------	-----------------	-----------------

The Overview tab summarises the event. An “About the Event” section shows the organizer’s description. An “Event Details” section shows:

- Team Size - the maximum number of members allowed per team
- Format - In-person or Virtual
- Visibility - Public or Private
- Allowed Technologies - shown as tags, if the organizer has restricted which technologies may be used

A “Prizes” section shows the 1st, 2nd, and 3rd place prize amounts configured for an event

6.3.2 Rules tab

← Go Home

AWS

[Download Certificate](#)
[Download Specifications](#)

Status	Prize Pool	Visibility	Format
Completed	R 0	Public	Virtual

[Overview](#) [Rules](#) [Announcements](#) [History](#) [Rankings](#)



NO RULES AVAILABLE

No rules have been provided for this event yet.

The Rules tab shows the full rules and requirements the organizer has written for the event. If the organizer hasn't added any rules yet, this tab shows "No Rules Available."

6.3.3 Announcements Tab

← Go Home

AWS

Download Certificate

Download Specifications

Status	Prize Pool	Visibility	Format
Completed	R O	Public	Virtual

Overview Rules Announcements History Rankings



NO ANNOUNCEMENTS

There are no announcements for this event yet.

The Announcements tab lists updates posted by the event's organizers, newest first. Each announcement shows its title, a severity badge (Info, Important, or Urgent, each with its own icon), the message, and when it was posted. If nothing has been posted yet, this tab shows "No Announcements."

6.3.4 Forum Tab

The Forum tab shows the event's discussion threads, so you can read what other participants and organizers are discussing and take part in the conversation for the event.

6.3.5 Making a Submission

The Submissions tab is where you upload your team's solution. You'll need to have joined or created a team first - if you haven't, this tab shows "You need to join a team" with a prompt to do so on the My Team tab. If the organizer hasn't published any levels yet, it shows "No levels available."

Levels are shown as their own set of tabs within the Submissions tab. Selecting a level shows:

- A description of the level and what it involves
- Important Notes - reminders that you may submit multiple times before the deadline (your highest score counts), that your solution must follow the submission requirements, and - for levels without a built-in coding environment - to download the level resources before getting started
- "Download Resources" - downloads the level's input files

For levels with a built-in coding environment, an "Open IDE" button takes you directly into it instead of a manual upload form. For other levels, a submission form asks for:

1. Solution Archive - your source code, as a .zip file
2. Output File - the output .json file your solution produced, which the official solver grades

Select "Submit Solution" once both files are attached. The button is disabled until both are provided, and shown "Submitting..." while your submission is queued for scoring by the official solver.

6.3.6 Submission History & Scoring Feedback

← Go Home

AWS

Download Certificate

Download Specifications

Status	Prize Pool	Visibility	Format
Completed	R 0	Public	Virtual

Overview Rules Announcements **History** Rankings

Submission History
View your submissions across different levels.

All Levels ▾ All Status ▾

Uploaded	Level	Status	Score	
26 Sept 2026, 16:46	Level 1	COMPLETED	80	Download Log
26 Sept 2026, 12:05	Level 1	COMPLETED	16.29	Download Log
26 Sept 2026, 12:04	Level 1	COMPLETED	80	Download Log

< 1 >

Showing 1 to 3 of 3 submissions

The Submission History tab lists every submission your team has made for the event, filterable by level and by status using the dropdown above the table. If you haven't submitted anything yet, this tab shows "No Submission History."

For each submission you can see its status as a colored badge, and select "Download Log" to download the detailed scoring log for that attempt - useful for understanding malformed output, rule violations, or other validation failures. Results are paginated when you have a lot of submissions.

6.3.7 Leaderboard

The Leaderboard tab shows live standings for the event. If you're on a team, a "My Team Performance" panel shows your team's name, current rank, and score, alongside a dropdown to switch which level's leaderboard you're viewing.

Below that, the full rankings table lists every team's rank (with gold/silver/bronze styling for the top three) and score. If the organizer hasn't made the leaderboard available yet, this tab shows "Leaderboard Currently Unavailable."

6.3.8 My Team

The My Team tab is where you create, join, and manage your team for the event. If you're not part of a team yet, this tab shows "You're not part of a team" with two options: "create Team" or "Request To Join".

Creating a Team

Select "Create Team", enter a unique Team Name, and confirm. You become the team's captain.

Requesting to Join a Team

Select "Request To Join" and enter the Join Code shared by an existing team's captain, then select "Send". The team's captain needs to approve your request before you're added.

Once You Have a Team

The "My Team" panel shows your team's name and four stats: number of Members, number of Submissions, total Score, and your Current Rank. Select "Leave Team" to leave (a confirmation dialog warns that you can only rejoin by submitting a new join request).

An "Invite to Your Team" panel shows your team's Join Code, with a "Copy" button to copy it and share with others. A "Team Members" table lists every member, tagging the captain as "Captain".

Managing Join Requests (Captains Only)

If you're the team's captain, a "Pending Requests" section lists anyone who asked to join, with their name and email. Select the check icon to approve a request or the cross icon to reject it. If nobody has requested to join, this shows "No pending requests."

See Part D : Help and Support for the in-platform Help Centre and frequently asked questions.

PART C

Authentication & Security

Both portals share the same underlying platform service:

- Role-Based Access Control: your account's role (Admin or Participant) determines which portal and features you can access
- Secure Sessions: the platform uses JWT-based authentication with secure session handling
- Content & File Storage: problem statements, level input files, solver versions, participant outputs, logs, branding assets, and source code archives are all stored and served securely by the platform.

PART D

Help and Support

[HOME](#) [HELP](#)[Help Center](#)

How can we help you?

Welcome to the help menu, find answers to frequently asked questions.

[Documentation](#)

User Guide

The complete user guide for organizers and participants.

[Open User Guide](#)[FAQ](#)

Frequently Asked Questions

Quick answers to the questions participants ask most often.

[How do I join an event?](#)[How do I create or join a team?](#)[How do I submit my solution?](#)[Can I submit multiple times?](#)[How are submissions graded?](#)[Where can I view my submission history?](#)

The in-platform Help Centre opens with the heading “How can we help you?” and the description “Welcome to the help menu, find answers to frequently asked questions.” It offers two ways to find answers: the complete User Guide, and a set of frequently asked questions.

1. User Guide

Under Documentation, select “Open User Guide” to open the complete user guide for organisers and participants - this manual.

2. Frequently Asked Questions

The FAQ section - described as “Quick answers to the questions participants ask most often” - is an expandable accordion covering:

- **How do I join an event?**
Browse the open events and select “Register” before the registration deadline; once accepted, the event appears in your Active Events Carousel
- **How do I create or join a team?**
Open the Teams section inside your event using the “View Team” button, then create a new team or join an existing one
- **How do I submit my solution?**
Select “Go to Event,” open the Upload Solution tab, upload your source archive and solution output, then submit before the event deadline
- **Can I submit multiple times?**
Yes- you can upload new submissions until the submission deadline; your latest valid submission for each level is used for the final results
- **How are submissions graded?**
The platform automatically evaluates your submissions using the organizer’s solver and generates a score along with a feedback log
- **Where can I view my submission history?**
Open your event and go to the Submission History tab to view your past submissions

PART E

Wow Factors

Beyond running events, registrations, and submissions, the platform includes three features that set it apart from a standard hackathon tool: automatic plagiarism detection, AI-assistance detection built on behavioural telemetry, and a built-in certificate designer and generator. All three are Admin-only.

7.1 AI Assistance Detection

For each team and level, the platform builds a Telemetry Report from the coding activity captured while a team works in the browser-based IDE - keystrokes, pastes, code runs, submissions, and time spent away from the tab - and uses it to estimate how likely it is that a submission was produced with AI assistance.

Reading the report

Use the “Select Level” tabs at the top of the report to switch between a team’s levels. Each level shows:

- Behavioural Risk Score - a score out of 100 with a review badge (Low, Medium, High, or Insufficient Data) and a progress bar
- Evidence Confidence - how much telemetry data was available to base the score on, as a percentage
- Triggered Indicators - the specific behavioral flags that contributed to the risk score
- AI Assistance Prediction - shown only when a prediction is available,, this displays the likelihood (as a percentage) that AI assistance was used, a “Flagged for Review” badge if it crosses the configured review threshold, and the threshold itself
- Telemetry Metrics - a detailed grid of the underlying activity data

Telemetry Metrics

The metrics grid includes:

- Events, Sessions, and Observed Duration
- Typed Characters, Pasted Characters, and Paste Fraction
- Large Pastes, and Paste/Large Paste After Return (pasting shortly after returning from another tab)
- Deleted Characters and Rework Ratio
- Runs, Successful Runs, and Code Errors
- Submissions
- Tab Away Time and Longest Tab Away
- Average Typing Interval

An “About AI Assistance Prediction” note on the report reminds you that the score comes from a machine learning model trained on telemetry patterns, that it estimates a likelihood rather than proves anything, and that it should be used as one additional signal alongside the behavioural risk score - not as a definitive determination on its own.

7.2 Plagiarism Checker

The Plagiarism Checker page compares every pair of submissions for an event to look for structural similarity between teams’ code, so you can catch copied or shared solutions before finalising results.

Running a check

1. Optionally filter to a single level using the level dropdown - leave it on “All levels” to check everything at once.
2. Select “Run Check” to start a new comparison run. A banner shows the run as Queued or Running, and updates automatically once it finishes; if a run fails, the banner shows the error so you can try again.
3. Once complete, results appear in a table listing every compared pair.

Reviewing results

Each row of the results table shows Team A, Team B, a structural Score(their similarity as a percentage), the number of Matched Fragments, and whether the pair is Flagged. Use the search box to find a specific team, and the “Showing flagged only” / “Showing all pairs” toggle to

focus on the pairs that need attention. Select “View Diff” on any pair to open a side-by-side comparison of the two submissions, so you can inspect exactly what matched.

If no check has been run yet, or the last run compared zero pairs (for example, because a level doesn’t have at least two scored submissions), the page explains why and prompts you to run a check.

7.3 Certificate Generator

The Certificate page lets you design a certificate template for an event and generate personalised, verifiable certificates for participants once the event has finished.

Designing a Template

A sidebar lists every template you’ve created for the event; select “+ New” to start another, or select an existing one to keep editing it. The designer itself has:

- A template name field, and a Background image upload for the certificate artwork
- A palette to add elements to the canvas: a Bound Field (text tied to participant data, such as their name or rank), a Static Label (fixed text), a QR Code (encodes a unique verification link for that certificate), or an Image/Signature
- A canvas where you drag elements into position

Selecting an element on the canvas opens its properties on the right:

- Text elements - choose static text or a bound field (for example, participant name or event name), plus font,size,colour, and alignment
- QR code elements - size only; it always encodes that certificate’s verification link
- Image element - replace the image, and sets its width and height
- Every element - its X/Y position on the canvas, and which certificate types include it, so, for example, a rank field can be hidden on participation certificates while still showing on winner certificates

Select “Save Template” once you’re happy with the design

Generating Certificates

Only trigger generation once the event has ended and the results are confirmed - if results change afterwards, you'll need to regenerate. Choose a generation scope:

- All participants - everyone who took part gets a certificate
- Top N Teams - only the top-ranked finalists, for a number of teams you specify

Select "Generate Certificates" (this requires a template to be selected first). A progress bar tracks the run's status and how many certificates have been completed out of the total, and shows an error message if generation fails.

Issued Certificates

A table at the bottom of the page lists every certificate issued for the event so far, showing the recipient, certificate type, their rank at the time of issue, when it was issued, and its unique verification code, with a "Download" link for each one. If none have been issued yet, this shows "No certificates issued for this event yet."